E-Commerce Sales & Customer Behavior Analysis

Exploratory Data Analysis (EDA) Report

Dataset: UCI Online Retail Dataset

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn

Analysis period: 1 December 2010 – 9 December 2011

Prepared from the accompanying Jupyter Notebook and its recorded outputs.

Purpose: transform transactional retail data into clear findings about revenue, products, customers, order behavior, and data quality.

1. Executive Summary

This project explores the UCI Online Retail transactional dataset to understand sales performance, product activity, customer contribution, order values, seasonality, and data-quality concerns. The original file contains 541,909 transaction lines and eight columns. After removing exact duplicates and rows without product descriptions, the cleaned working dataset contains 535,187 rows. A separate positive-sales subset contains 524,878 rows and is used for most sales analyses.

The analysis shows that the United Kingdom is the largest revenue market in the dataset, revenue rises sharply from September through November 2011, and a small number of products and customers account for substantial recorded revenue. Transaction quantities, prices, and line revenue are strongly right-skewed, so extreme values materially affect averages and charts.

Headline metrics

Metric	Result
Original dataset	541,909 rows × 8 columns
Cleaned working dataset	535,187 rows
Positive-sales analysis subset	524,878 rows
Invoices in positive-sales subset	19,960
Unique products (StockCode)	3,922
Countries represented in positive-sales subset	38
Total recorded line revenue in positive-sales subset	10,642,110.80
Average order value (mean)	533.17
Median order value	303.30

Interpretation note: monetary values are reported in the dataset's monetary units. The revenue total includes postage and other service/manual entries that appear in the transaction descriptions; it should not be described as merchandise-only revenue.

2. Project Objectives

- Inspect the dataset's structure, fields, data types, missing values, duplicates, cancellations, and unusual numeric values.
- Clean the data sufficiently for exploratory analysis while retaining rows with missing customer IDs where possible.
- Compare revenue and quantity across products and countries, and examine monthly trends.
- Explore order value, customer contribution, transaction distributions, and relationships between numerical variables.
- Translate the observed patterns into practical business recommendations and state analytical limitations.

3. Dataset Overview

The dataset records individual retail invoice lines. Its eight fields are InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. Because each row is a line item rather than necessarily a complete order, line-level counts and invoice-level metrics should not be mixed without aggregation.

The recorded period runs from 1 December 2010 at 08:26 to 9 December 2011 at 12:50. December 2011 is therefore only a partial month, which matters when comparing monthly revenue and quantity.

4. Data Quality and Cleaning

Check / action	Observed result or treatment
Missing Description	1,454 rows initially; rows without descriptions removed
Missing CustomerID	135,080 initially; 133,583 remain after cleaning; retained for analyses that do not require customer identity
Exact duplicate rows	5,268 duplicates removed
Cancellation invoice lines	9,288 rows initially identified by invoice numbers beginning with “C”
Negative quantities	10,624 initially; 9,725 remain in the cleaned working dataset
Non-positive prices	1,058 rows in the final cleaning checks (including zero and negative prices)
Positive-sales subset	Excludes cancellation invoices and rows with Quantity ≤ 0 or UnitPrice ≤ 0
Derived field	Revenue = Quantity × UnitPrice

The notebook preserves the original dataframe by creating a cleaned copy. It removes exact duplicates and missing product descriptions, converts InvoiceDate and numeric fields to appropriate types, and creates a Revenue field. A separate df_sales subset is then used for most commercial analyses.

Important limitation: the positive-sales subset is a practical analysis filter, not a complete accounting reconciliation. The retained cleaned dataframe still includes some negative-quantity and non-positive-price records, and customer IDs remain missing for many rows. Findings should be interpreted as exploratory.

5. Revenue and Product Performance

The highest recorded revenue by product description includes postage and service/manual entries, so the ranking reflects recorded transaction-line revenue rather than a clean list of physical merchandise alone.

Product / description	Recorded revenue
DOTCOM POSTAGE	206,248.77
REGENCY CAKESTAND 3 TIER	174,156.54
PAPER CRAFT, LITTLE BIRDIE	168,469.60
WHITE HANGING HEART T-LIGHT HOLDER	104,284.24
PARTY BUNTING	99,445.23
JUMBO BAG RED RETROSPOT	94,159.81
MEDIUM CERAMIC TOP STORAGE JAR	81,700.92
POSTAGE	78,101.88

Product / description	Recorded revenue
Manual	77,750.27
RABBIT NIGHT LIGHT	66,870.03

By quantity, PAPER CRAFT, LITTLE BIRDIE recorded 80,995 units and MEDIUM CERAMIC TOP STORAGE JAR recorded 78,033 units. These exceptionally large totals should be treated as investigation candidates: bulk orders, unusual transactions, or data-specific events can dominate quantity rankings.

The most frequently recorded product line was WHITE HANGING HEART T-LIGHT HOLDER (2,311 transaction lines), followed by JUMBO BAG RED RETROSPOT (2,109) and REGENCY CAKESTAND 3 TIER (2,007). Frequency, quantity, and revenue measure different types of product performance and should be reported separately.

6. Country Performance

Country	Recorded revenue
United Kingdom	9,001,744.09
Netherlands	285,446.34
EIRE	283,140.52
Germany	228,678.40
France	209,625.37
Australia	138,453.81
Spain	61,558.56
Switzerland	57,067.60
Belgium	41,196.34
Sweden	38,367.83

The United Kingdom contributes approximately 9.00 million in recorded revenue, substantially more than any other country in the top ten. This indicates a strong concentration in the home market in this dataset. International expansion opportunities should be evaluated with order counts, customer counts, shipping costs, margins, and market size—not revenue alone.

7. Monthly Revenue and Demand

Monthly revenue fluctuates during the first part of 2011 and rises substantially from September. The highest recorded monthly revenue is November 2011 at approximately 1.50 million. Monthly quantity also rises from September and peaks in November. December 2011 declines, but the dataset ends on 9 December, so that month is incomplete and should not be compared directly with full months.

Month	Revenue
Dec 2010	821,452.73
Jan 2011	689,811.61
Feb 2011	522,545.56
Mar 2011	716,215.26
Apr 2011	536,968.49
May 2011	769,296.61
Jun 2011	760,547.01
Jul 2011	718,076.12
Aug 2011	757,841.38
Sep 2011	1,056,435.19
Oct 2011	1,151,263.73
Nov 2011	1,503,866.78

Month	Revenue
Dec 2011 (partial)	637,790.33

The late-year uplift may be consistent with seasonal or holiday-related demand, but the notebook's descriptive analysis does not establish the cause. Comparing with other years or checking product-level and week-level patterns would help validate seasonality.

8. Order Value and Customer Contribution

The positive-sales subset contains 19,960 invoices. Average order value is 533.17, while median order value is 303.30. Because the mean is substantially higher than the median, the order-value distribution is likely right-skewed: a smaller number of large orders pull the average upward. Reporting both values gives a more balanced view.

Customer ID	Recorded revenue
14646	280,206.02
18102	259,657.30
17450	194,390.79
16446	168,472.50
14911	143,711.17
12415	124,914.53
14156	117,210.08
17511	91,062.38
16029	80,850.84
12346	77,183.60

Customer 14646 is the largest recorded contributor among customers with an available CustomerID. These customers may be candidates for retention or account-management analysis, but the ranking alone does not establish profitability, loyalty, or future value.

9. Distributions, Outliers, and Relationships

The quantity, unit-price, and line-revenue distributions are highly skewed, with a small number of extreme values. The notebook uses histograms and box plots to examine these distributions and uses 99th-percentile limits to make the main body of the scatter plot easier to inspect.

Variable	Key distribution result
Quantity	Median 4; 75th percentile 11; maximum 80,995
UnitPrice	Median 2.08; 75th percentile 4.13; maximum 13,541.33
Line revenue	99th percentile 183.60
99th-percentile quantity limit	100 units
99th-percentile unit-price limit	16.98

The largest recorded quantity transaction is 80,995 units of PAPER CRAFT, LITTLE BIRDIE, with a recorded line revenue of 168,469.60. The highest unit-price records include AMAZON FEE, Adjust bad debt, POSTAGE, DOTCOM POSTAGE, and Manual. These entries should be reviewed separately from ordinary merchandise before making product-pricing conclusions.

The Pearson correlation between UnitPrice and Quantity is –0.0038, which indicates virtually no linear association across the selected rows. Quantity and Revenue show a strong positive correlation of approximately 0.91; however, Revenue is calculated as Quantity × UnitPrice, so this relationship is partly built into the definition of the variable and is not evidence of an independent causal relationship.

Visualisations included in the notebook

- Bar charts: leading products by revenue, countries by revenue, products by quantity, most frequently recorded products, and leading customers by revenue.
- Time-series charts: monthly revenue, monthly quantity, and monthly average order value.
- Distribution plots: histograms for quantity, unit price, and line revenue; box plots for quantity, unit price, and line revenue.
- Relationship plots: unit price versus quantity scatter plots, including a view trimmed at the 99th-percentile display limits, and a correlation heatmap.
- Count/category plots: cancellation invoice counts and transaction-line counts for the top countries, plus a keyword-based product grouping chart.

The keyword-based product grouping is approximate because it infers categories from product descriptions rather than using an official category field. It should be treated as a rough exploratory view, not a definitive product taxonomy.

10. Business Recommendations

- **Plan for the late-year peak.** Review stock availability, fulfilment capacity, staffing, and promotional timing ahead of the September–November uplift. Validate this pattern against additional years before treating it as a recurring seasonal effect.
- **Separate merchandise from service entries.** Flag postage, manual adjustments, fees, and other non-merchandise descriptions before publishing product rankings or merchandise-only revenue.
- **Investigate extreme transactions.** Review the largest quantities and prices for bulk orders, unusual invoice lines, and data-quality issues. Preserve legitimate wholesale orders rather than deleting outliers automatically.
- **Monitor customer concentration.** Track high-revenue customers and consider retention or account-management actions, while also checking margins, repeat frequency, and customer-level missingness.

- **Use multiple order-value measures.** Report median and mean order value together, and segment order values by country, month, and customer type where reliable fields are available.
- **Improve customer data capture.** Since many rows have missing CustomerID values, investigate the source of missing IDs and report the coverage of customer-level analyses.
- **Evaluate markets with context.** Use revenue alongside customer counts, invoice counts, margin, fulfilment cost, and growth to assess international opportunities.

11. Limitations and Further Work

- The analysis is descriptive and does not establish causal explanations for changes in revenue, quantity, or customer behavior.
- December 2011 is incomplete, ending on 9 December; monthly comparisons should explicitly account for this.
- The dataset contains missing CustomerID values, unusual transaction sizes, service entries, and other special records that affect interpretation.
- The revenue field is line quantity multiplied by unit price; it is not necessarily profit, cash collected, or merchandise-only sales.
- The cancellation percentage is calculated as 3,836 cancelled invoice numbers out of 24,446 unique invoice numbers (15.69%). This is an invoice-level share, not a share of line items or revenue.
- The positive-sales subset excludes cancellation invoices, non-positive quantities, and non-positive prices. It is suitable for the notebook's exploratory sales comparisons but is not an accounting reconciliation.
- Product categories inferred from description keywords are approximate and can misclassify items.

Recommended next steps

- Create separate merchandise, postage, fee, and manual-adjustment flags and recompute product-level rankings.
- Calculate cancellation rates at invoice, line-item, and value levels using clearly defined denominators.
- Add repeat-purchase frequency, customer recency, and customer monetary value for customers with reliable IDs.
- Compare full months only, or normalize partial December revenue by days observed with an explicit caveat.
- Where available, calculate gross margin and shipping costs to distinguish revenue from business profitability.

12. Conclusion

This EDA project demonstrates a complete exploratory workflow: loading and inspecting a real-world dataset, identifying and treating data-quality issues, deriving a revenue measure, comparing product and country performance, analysing monthly trends and order values, and using visualisations to investigate distributions and relationships.

The clearest findings are the dominance of the United Kingdom in recorded revenue, a substantial rise in revenue and quantity during September–November 2011, a right-skewed order-value distribution, and a small set of products and customers with large recorded contributions. These findings can inform stock planning, customer retention, and further market analysis, provided that service entries, extreme transactions, missing customer IDs, and the partial final month are handled transparently.

Dataset source: UCI Machine Learning Repository — Online Retail Dataset (ID 352). The report is based on the accompanying notebook's recorded outputs; all values should be rechecked if the notebook or cleaning rules change.